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$$\int x\sqrt{1-4x^2}dx$$

$$\text{Stel } t = 1 - 4x^2 \Rightarrow dt = -8xdx$$

$$\Rightarrow x = \frac{\sqrt{t-1}}{4} \Rightarrow dx = \frac{dt}{-8x}$$

$$= \int \frac{\sqrt{\frac{t-1}{-4}}\sqrt{t}}{-8\sqrt{\frac{t-1}{-4}}}dt$$

$$= \int \frac{\sqrt{t}}{-8}dt$$

$$= -\frac{1}{8} \int t^{\frac{1}{2}}dt$$

$$= -\frac{1}{8} \frac{t^{\frac{3}{2}}}{\frac{3}{2}} + C$$

$$= -\frac{\sqrt{t^3}}{12} + C$$

$$= -\frac{\sqrt{(1-4x^2)^3}}{12} + C$$

References